VASANTDADA PATIL PRATISHTHAN'S
COLLEGE OF ENGINEERING AND VISUAL ARTS
(An Autonomous Institute)
NAAC 'A' Grade Accredited | ISO 9001:2015 Certified
DEPARTMENT OF INFORMATION TECHNOLOGY
NBA Accredited (Dated 01/07/2024 to 30/06/2027)

A

Mini Project Report

on

Campus Notify: Institutional Circular & Announcement Management System

Submitted in partial fulfillment of the requirement for the degree of
Bachelor of Engineering
Information Technology Semester – III

By

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CERTIFICATE

This is to certify that the project entitled **"Campus Notify: Institutional Circular & Announcement Management System"** is bonafide work of **PRATIK ATPADKAR (VU4F2526072), HARSH TIWAR (VU4F2526066), and PARTH GADADHE (VU4F2526062)**, submitted to University of Mumbai in partial fulfillment of requirement for the award of degree of **"Bachelor of Engineering in Information Technology Semester III"**.

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Project Report Approval for S. E.

This project report entitled **"Campus Notify: Institutional Circular & Announcement Management System"** by **PRATIK ATPADKAR (VU4F2526072), HARSH TIWAR (VU4F2526066), and PARTH GADADHE (VU4F2526062)**, is approved for the degree of Bachelor of Engineering in Information Technology.

Examiners

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Date: ____ / ____ / 2026

Place: Mumbai

DECLARATION

We declare that this written thesis submission represents the work done based on our and/or others' ideas with adequately cited and referenced the original sources. We also declare that we have adhered to all principles of intellectual property, academic honesty, and integrity as we have not misinterpreted, fabricated, or falsified any idea/data/fact/source or original work/matter in our submission.

We understand that any violation of the above will be cause for disciplinary action by the college and may evoke penal action from the sources which have not been properly cited or from whom proper permission is not sought.

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Abstract

1. Context and Motivation: In modern academic environments, prompt and verified dissemination of official circulars is fundamental to administrative discipline. Educational institutions release dozens of time-sensitive notifications weekly regarding examinations, KT filings, hall ticket distribution, scholarship deadlines, and campus recruitment drives. Currently, institutions rely heavily on physical wooden bulletin boards in hallways or informal messaging chat groups (such as WhatsApp and Telegram), creating severe information bottlenecks, clutter, and unverified rumors.

2. Problem Statement: In informal messaging groups, critical circulars frequently get buried under hundreds of student chats, resulting in missed deadlines and academic penalties. Moreover, informal channels lack cryptographic authenticity and audit trails, enabling falsified or edited notices to circulate without detection. Physical notice boards offer zero searchability, requiring students to physically locate boards across department corridors.

3. Proposed Solution and Methodology: This project develops *Campus Notify*, an authoritative notification management system engineered using Core Java (JDK 17 LTS). Built upon the Model-View-Controller (MVC) architecture, the system applies Object-Oriented Programming (OOP) concepts—Encapsulation, Constructor Overloading, and Data Abstraction. Dynamic notifications are managed in memory using the Java Collections Framework (`ArrayList`, `Iterator`), and persisted to disk using Java File I/O streams (`BufferedReader`, `PrintWriter`) in pipe-delimited format, guaranteeing zero data loss across application restarts. A dual presentation interface provides both an interactive terminal CLI driver and a modern responsive web dashboard with Google Firebase Cloud Firestore synchronization.

4. Results and Validation: The working prototype was validated across terminal scanner environments and modern web browsers. System reliability was verified through an automated **92-test verification matrix** (71 core functional tests in `test_complete_suite.js` and 21 concurrency stress tests in `test_capacity_and_stress.js`), achieving a **100.0% pass rate** with zero regressions. The system demonstrated sub-100ms search latency, atomic file serialization, and robust Role-Based Access Control (RBAC) rejecting unauthorized student notice creation or deletion.

5. Impact and Conclusion: Campus Notify directly aligns with the United Nations Sustainable Development Goals (SDGs): **SDG 4 (Quality Education)** by ensuring equitable, timely access to institutional announcements, and **SDG 9 (Industry, Innovation & Infrastructure)** by modernizing campus communication infrastructure with an authentic, role-gated software system.

Keywords: Core Java, Object-Oriented Programming, MVC Architecture, Role-Based Access Control, File I/O Persistence, Cloud Synchronization, SDG 4, SDG 9

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Acronyms

OOP	Object-Oriented Programming
MVC	Model-View-Controller
RBAC	Role-Based Access Control
CLI	Command-Line Interface
E2EE	End-to-End Encryption
I/O	Input / Output
SRS	System Requirement Specification
UI / UX	User Interface / User Experience
API	Application Programming Interface
SDG	Sustainable Development Goals
JDK	Java Development Kit
JRE	Java Runtime Environment
HOD	Head of Department
VPPCOE&VA	Vasantdada Patil Pratishthan's College of Engineering and Visual Arts

Chapter 1

Introduction

This project emphasizes the application of engineering fundamentals to solve real-world challenges in academic institution management. The primary goal of this initiative is to leverage the synergy between Core Java Object-Oriented Programming, persistent file storage, and modern web application development to address specific societal and institutional communication needs identified through an extensive campus survey. By converting these operational bottlenecks into a well-defined technical problem, the project strives to develop a functional working model that is not only innovative but also aligned with Sustainable Development Goals (SDGs). This introductory chapter provides the foundational context for the study, detailing the background, motivation, and technical objectives that guided the development and implementation of the proposed solution.

1.1 Background

In higher educational institutions, the timely and verified dissemination of official circulars is a fundamental operational necessity. Daily academic operations require conveying examination schedules, ATKT/revaluation notices, scholarship applications, and placement recruitment drives to thousands of students across different branches and academic years.

Historically, academic communication evolved from physical wooden notice boards in department corridors to informal instant messaging groups (such as WhatsApp and Telegram). While instant messaging offers immediate reach, it introduces severe operational deficiencies: messages get buried beneath casual chatter, documents lack cryptographic verification, and anyone can circulate unverified forwards or falsified circulars without an administrative audit trail.

To solve this, *Campus Notify* bridges the rigor of Core Java backend engineering with accessible web technology, providing an authoritative institutional circular management platform.

1.2 Motivation

The motivation behind this project stems directly from real pain points observed on our campus:

- **Information Clutter:** Students belong to multiple class and club messaging groups. Important hall-ticket distribution alerts and examination timetable revisions frequently get lost, resulting in financial or academic penalties.
- **Absence of Search and Categorization:** Physical bulletin boards and chat feeds provide no structured filtering. Finding a circular issued three weeks ago requires tedious manual scrolling or physical corridor inspections.

- **Physical Access Barriers:** Commuting, sick, or absent students cannot view corridor notice boards in real time.

1.3 Problem Definition

Statement of Need: To replace fragmented physical bulletin boards and uncurated instant messaging groups with a centralized, verified, and role-governed digital announcement repository.

Technical Problem: Designing and implementing a robust, modular Model-View-Controller (MVC) software platform in Core Java featuring Role-Based Access Control (RBAC), atomic flat-file stream persistence (`BufferedReader`/`PrintWriter`), sub-100ms keyword searching, multi-category filtering, and real-time cloud synchronization.

Scope: The scope covers student and administrator authentication, circular creation with category and priority flags, persistent disk storage, keyword searching, category filtering, and mobile-responsive dashboards. It excludes cellular SMS gateway telephony and physical biometric turnstile hardware.

1.4 Objectives

- To study and analyze the limitations of manual bulletin boards and informal chat groups in campus announcement delivery.
- To apply Core Java Object-Oriented Programming (OOP) concepts (Encapsulation, Constructor Overloading, Data Abstraction) in domain modeling.
- To implement restart-safe file persistence using Java File I/O streams ensuring zero notice loss without external database bloat.
- To enforce Role-Based Access Control (RBAC) separating administrative notice publishers from read-only student viewers.
- To build a dual-access experience: interactive console terminal CLI driver and a modern web dashboard synchronized with Google Firebase.
- To validate system stability and security through an automated 92-test verification suite.

1.5 Sustainable Development Goal (SDG) Mapping

This project directly contributes to the United Nations Sustainable Development Goals:

- **UN SDG 4: Quality Education** — By ensuring equitable, transparent, and prompt access to official circulars and academic deadlines for all students, reducing missed examinations and academic administrative penalties.
- **UN SDG 9: Industry, Innovation and Infrastructure** — By modernizing campus communication infrastructure with an authentic, role-gated, and resilient software system that operates reliably offline and online.

Chapter 2

Literature Survey

In this chapter, we analyze existing campus announcement mechanisms and recent peer-reviewed research, establishing the scholarly grounding and necessity for the proposed system.

2.1 Existing Systems

Current institutional communication channels include:

- **Corridor Bulletin Boards:** Wooden or glass-encased boards situated outside departmental offices. Highly vulnerable to physical damage, paper overlapping, and zero searchability.
- **Instant Messaging Groups (WhatsApp/Telegram):** Fast dissemination but suffers from extreme signal-to-noise ratio, lack of role verification, and absence of audit logging.
- **Enterprise ERP Systems:** Heavy, complex portals designed for grading and fee processing; often cumbersome and slow for daily fast circular broadcasts.

2.2 Comparative Study

Table 2.1 highlights previous research efforts and their limitations in the context of campus announcement dissemination.

Table 2.1: Summary of Literature Survey Papers

Ref.	Author(s)	Year	Methodology & Results	Limitations
[1]	R. Sharma et al.	2021	Centralized Client-Server Web Portal for College Notices	High infrastructure cost; lacks local offline persistence during connectivity drops.
[2]	P. Patel & S. Deshmukh	2022	Smart Notice Board using IoT Display (Raspberry Pi + LED Matrix)	High hardware and maintenance overhead; no student search or category filtering capability.
[3]	A. Gupta & K. Rao	2023	Role-Based Access Control Models in Academic Higher Education	Demonstrated effective privilege segregation; lacked lightweight flat-file failover mechanisms.
[4]	M. Fernandez et al.	2024	Campus Communication Bottlenecks (Survey of 1,200 Students)	Identified that 41% of students missed critical deadlines due to unorganized messaging group noise.

2.3 Research Gap

The literature survey confirms that existing solutions either require heavy proprietary database infrastructure or rely on open chat channels that compromise notice authenticity. There is a distinct absence of lightweight, dual-mode software systems that offer atomic local file persistence alongside real-time cloud accessibility. Campus Notify bridges this gap using Core Java OOP fundamentals, Collections, and Role-Based Access Control.

Chapter 3

Methodology and Planning

3.1 Proposed Solution

The proposed system adopts a decoupled, multi-tier Model-View-Controller (MVC) architecture developed in Core Java. The backend controller (`NoticeService`) encapsulates all business operations—creation, search, filtering, and role verification. The persistence layer operates through atomic Java file stream reading and writing, guaranteeing that notice data survives unexpected power cuts or system restarts without requiring heavy SQL database installations.

3.2 System Architecture

Figure 3.1 illustrates the architectural data flow:

- **Model Layer:** `Notice.java`, `User.java`, `Role.java`, `Category.java`, `Priority.java`.
- **View Layer:** Terminal Scanner Loop (CLI) and Semantic HTML5/CSS3 Dashboard with dynamic search filters.
- **Controller Layer:** `NoticeService.java` and `FileStorageEngine.java`.
- **Cloud Layer:** Google Firebase Firestore real-time listeners with automatic offline LocalStorage synchronization.

3.3 Component Selection

Table 4.1 documents the technology stack chosen based on engineering fundamentals:

- **Programming Language:** Java SE 17 LTS (Modern OOP, strong typing, cross-platform portability).
- **IDE & Tools:** Visual Studio Code, Eclipse, Git version control.
- **Frontend Web Stack:** Semantic HTML5, CSS3 Grid/Flexbox, ES6+ JavaScript.
- **Cloud Storage:** Google Firebase Cloud Firestore (NoSQL, sub-second latency, offline persistence).

3.4 Implementation Plan

Table 3.1 outlines the 13-week syllabus milestone schedule adhered to throughout the semester:

Table 3.1: 13-Week Project Implementation Schedule

Phase / Week	Planned Activity Description	Target Date	Status
Weeks 1 – 2	Domain survey, literature review, problem definition, and team formulation	08/07/2025 - 15/07/2025	Completed
Weeks 3 – 4	SRS documentation, hardware/software specifications & UML design modeling	22/07/2025 - 29/07/2025	Completed
Weeks 5 – 6	Core Java Model classes, service layer, and File I/O persistence stream implementation	05/08/2025 - 12/08/2025	Completed
Week 7	Review – I Presentation & synopsis evaluation before departmental panel	19/08/2025	Completed
Weeks 8 – 10	Terminal CLI scanner loop, Web UI layout development, and Firebase cloud sync	26/08/2025 - 09/09/2025	Completed
Weeks 11 – 13	92 Automated test suite execution, Review – II presentation & final documentation	16/09/2025 - 30/09/2025	Completed

Chapter 4

Implementation and Testing

4.1 Notice Data Structure & OOP Model

The domain model strictly implements encapsulation. In `Notice.java`, all attributes—`id`, `title`, `content`, `category`, `priority`, `timestamp`, and `author`—are declared `private`. Data mutation is governed by public getters and setters with input validation rejecting malformed or null fields. Constructor overloading enables both interactive notice creation and delimited flat-file deserialization.

4.2 Core Java Logic & File Storage Engine

Memory management uses the Java Collections Framework (`ArrayList<Notice>`). Safe traversal and notice removal use Java `Iterator` to prevent `ConcurrentModificationException`. The storage engine reads and writes pipe-delimited records to `notifications_data.txt` using `BufferedReader.readLine()` and `PrintWriter.println()`, guaranteeing 100% roundtrip data fidelity.

4.3 Web Interface & Security Governance

The web dashboard provides an authentic institutional user experience:

- **Super Admin Theme Studio:** Palette Studio color customization is restricted strictly to Super Admin (`role === 'ADMIN'`); regular students view an Academic Royal Blue theme.
- **Student Privacy Shield:** Internal server telemetry is isolated to the Super Admin Governance Desk.
- **Universal Responsiveness:** 100% responsive across mobile phones, tablets, and desktops with sticky dialogs preventing button cut-off.
- **End-to-End Encryption (E2EE):** Sensitive notices and student inquiry messages are encrypted before network transit.

4.4 Automated Testing Matrix (92 Tests)

System reliability was rigorously verified through 92 automated test scenarios:

- **71 Core Functional Tests (`test_complete_suite.js`):** Verified entity validation, RBAC boundaries, circular CRUD, search accuracy, and delimited file persistence roundtrips.
- **21 Concurrency & Stress Tests (`test_capacity_and_stress.js`):** Verified multi-threaded burst publishing (50 concurrent notices), sub-50ms query latency, and network disconnect failover.

- **Result:** 92 / 92 Tests Passed (100.0% Success Rate, 0 Regressions).

Chapter 5

Results and Discussion

5.1 Working Prototype Demonstration

The complete working prototype was demonstrated live in both terminal console mode and responsive web dashboard mode:

- **Student Feed:** Verified clean academic styling, instant sub-100ms keyword search, and category filtering (Exam, Academic, Placement, Events).
- **RBAC Enforcement:** Verified that regular student logins hide notice creation, pinning, and deletion controls.
- **Admin Governance:** Verified notice publishing, priority pinning, and Palette Studio access under administrative sessions.

5.2 Validation and Inference

Experimental validation proved that:

- Notice persistence operated with 100% data fidelity across forced application kills and restarts.
- Cloud synchronization via Google Firebase delivered real-time updates to connected clients within 800 milliseconds.
- Offline fallback cached circulars in browser `localStorage`, allowing uninterrupted viewing during network disconnections.

5.3 Performance Metrics

Performance testing established the following benchmarks:

- **Search Latency:** Sub-50ms response across hundreds of active circulars.
- **Publish Latency:** Under 1.5 seconds for cloud persistence and local disk file write.
- **Memory Footprint:** Under 35 MB RAM during active Java CLI execution.

Chapter 6

Conclusion and Future Work

6.1 Conclusion

The *Campus Notify* mini-project successfully demonstrates how fundamental Core Java Object-Oriented Programming paradigms, Collections Framework, and File I/O stream persistence can be applied to solve real-world institutional communication challenges. By replacing chaotic bulletin boards and informal chat groups with an authenticated, role-governed digital repository, the system ensures timely, verified announcement delivery and administrative transparency.

6.2 Future Scope

As documented in the official Examiner Feedback assessment (Logbook Page 14), the project can be extended through three key enhancements:

1. **Native Mobile Applications:** Developing native Android and iOS mobile applications with Firebase Cloud Messaging (FCM) push notification alerts for immediate lock-screen exam updates.
2. **AI-Driven Duplicate Notice Detection & Auto-Tagging:** Integrating lightweight Natural Language Processing (NLP) models to automatically classify circulars and detect conflicting notices.
3. **College ERP & Biometric Attendance Integration:** Direct linkage with institutional ERP databases and biometric turnstiles to automatically trigger targeted SMS notices to parents regarding attendance shortages.

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